

```

short_window = 20
long_window = 50
df["SMA20"] = (
    df["Close"]
    .rolling(short_window)
    .mean()
)
df["SMA50"] = (
    df["Close"]
    .rolling(long_window)
    .mean()
)
print(
    df[
        ["Close", "SMA20", "SMA50"]
    ].tail()
)

```

Close	SMA20	SMA50
3892.482	4042.155	3910.172
3882.796	4034.544	3910.964
3878.838	4030.816	3912.046
3873.203	4025.599	3913.421
3813.598	4017.13	3913.343

Plotting the moving averages

Though browsing the record set with a short- and long-term moving average shows new feature columns,

visualisation plays a special role in understanding the price movement pattern.

```

plt.figure(figsize=(15, 7))
plt.plot(
    df.index,
    df["Close"],
    label="Close Price",
    linewidth=1.2
)
plt.plot(
    df.index,
    df["SMA20"],
    label="20-Day SMA",
    linewidth=2
)
plt.plot(
    df.index,
    df["SMA50"],
    label="50-Day SMA",
    linewidth=2
)
plt.title(
    f"{ticker} - Moving Average Crossover"
)

plt.xlabel("Date")
plt.ylabel("Price")
plt.legend()
plt.grid(True, alpha=0.3)
plt.tight_layout()
plt.show()

```

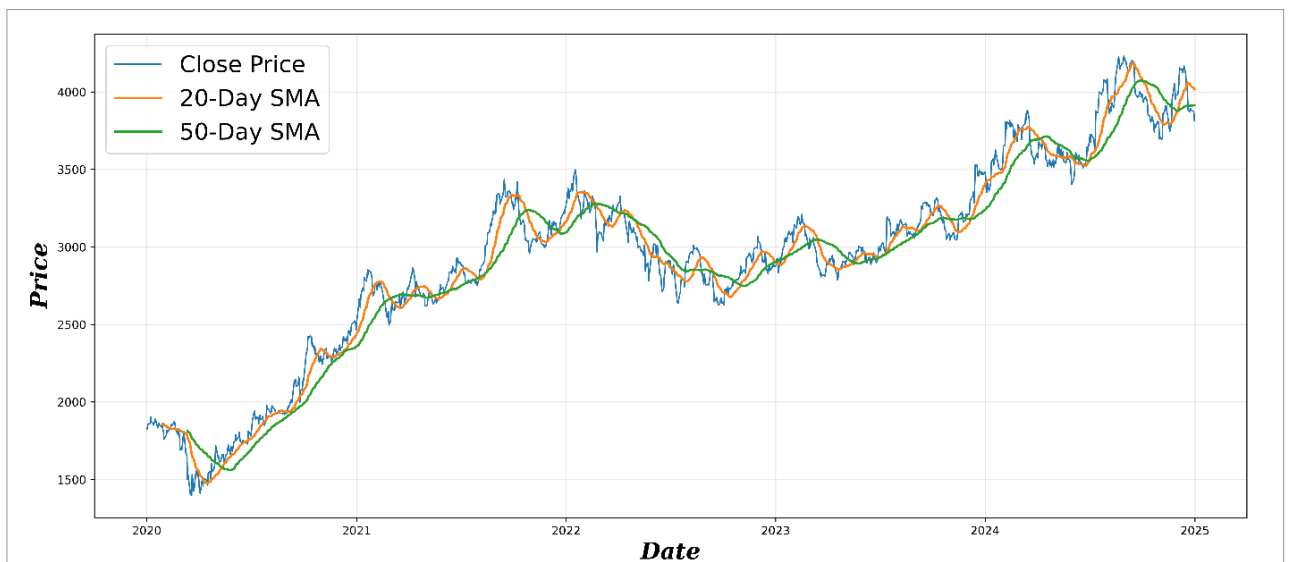


Figure 2: Moving average crossover for a 20/50-day period